

EDA (Exploratory Data Analysis)

- What is Exploratory Data Analysis?
 - Purpose of EDA in the data analysis process
 - Importance of visualizing data
- Tools and Libraries for EDA
 - Python libraries: Pandas, Matplotlib, Seaborn, Plotly, and Statsmodels
 - Overview of Jupyter Notebooks and DataFrames
 - Basics of DataFrames and Series (Pandas)

2. Understanding Data

2.1 Data Collection

- Types of Data: Qualitative vs Quantitative
- Sources of Data: Structured, Semi-structured, Unstructured
- Types of Variables: Continuous, Discrete, Nominal, Ordinal

2.2 Data Loading and Inspection

- Loading Data into Pandas DataFrame (CSV, Excel, SQL, JSON)
- Overview of DataFrames: Columns, Rows, Indexing
- Checking Basic Information
 - `df.info()`, `df.describe()`, `df.shape`, `df.head()`
- Handling Large Datasets: Sampling, Chunking

2.3 Data Quality and Missing Data

- Identifying Missing Data: `df.isnull()`, `df.isna()`
- Handling Missing Values:
 - Dropping or Imputing Missing Data (Mean, Median, Mode, Interpolation)
 - Using `SimpleImputer` from Scikit-learn
- Identifying Duplicates and Removing Them
 - `df.duplicated()` and `df.drop_duplicates()`
- Handling Inconsistent Data: Text standardization, Date parsing issues

3. Data Cleaning

3.1 Data Transformation

- Data Type Conversion (`df.astype()`)
- Encoding Categorical Data (Label Encoding, One-Hot Encoding)
- DateTime Parsing and Transformation
- Handling Outliers: Z-Score, IQR, Winsorizing

3.2 Data Normalization and Scaling

- Min-Max Scaling and Standardization
- Log Transformation and Power Transformation
- Using Scikit-learn's `StandardScaler` and `MinMaxScaler`

4. Univariate Analysis

4.1 Descriptive Statistics

- Measures of Central Tendency: Mean, Median, Mode
- Measures of Dispersion: Range, Variance, Standard Deviation, Quartiles, IQR
- Frequency Distribution
 - Using `df.value_counts()`, `df.groupby()`

4.2 Distribution and Visualization

- Understanding the Distribution of Data
 - Histograms (`plt.hist()`)
 - Box Plots (`sns.boxplot()`)
 - KDE (Kernel Density Estimation) Plots (`sns.kdeplot()`)
 - Violin Plots for Univariate Data Distribution
- Identifying Skewness and Kurtosis
 - Using `scipy.stats.skew()` and `scipy.stats.kurtosis()`

5. Bivariate Analysis

5.1 Correlation Analysis

- Understanding Correlation: Positive, Negative, and Zero Correlation
- Pearson and Spearman Correlation Coefficients
- Visualizing Correlations with Heatmaps
 - Using `sns.heatmap()` and `sns.pairplot()`
- Scatter Plots and Pair Plots

5.2 Categorical Data Analysis

- Exploring Relationships Between Categorical Variables
 - Cross-tabulation with `pd.crosstab()`
 - Chi-square Test for Association
 - Stacked Bar Plots and Count Plots (`sns.countplot()`)

5.3 Visualizing Relationships

- Scatter Plots and Line Plots for Continuous Variables
- Joint Plots and Pair Plots for Multivariate Analysis (`sns.jointplot()`)
- Heatmaps for Correlation Matrices

6. Multivariate Analysis

6.1 Multivariate Visualization

- Pairwise Plot and Matrix Plots
 - `sns.pairplot()` and `sns.pairgrid()`
- Grouped Bar Plots, Stacked Area Plots, and Facet Grids
- 3D Scatter Plots and Surface Plots

6.2 Principal Component Analysis (PCA)

- Introduction to Dimensionality Reduction
- Performing PCA on Multivariate Data using Scikit-learn
- Visualizing Reduced Dimensions
- Understanding Variance Explained by Principal Components

7. Feature Engineering

7.1 Feature Extraction

- Extracting Features from DateTime (Year, Month, Day, Weekday)
- Feature Creation from Text: TF-IDF, Word Embeddings
- Feature Creation for Time Series Data: Lagging, Rolling Window Statistics

7.2 Feature Selection

- Correlation Matrix and Heatmap for Feature Selection
- Recursive Feature Elimination (RFE)
- Feature Importance using Tree-based Models (Random Forest, XGBoost)

7.3 Handling Imbalanced Data

- Techniques for Dealing with Imbalanced Data: Resampling, Synthetic Data Generation (SMOTE)
- Using `imblearn` for balancing classes

8. Statistical Analysis

8.1 Hypothesis Testing

- Null and Alternative Hypothesis
- Types of Errors (Type I and Type II)
- T-tests, Z-tests, and ANOVA
- Chi-Square Test for Independence

8.2 Confidence Intervals

- Understanding Confidence Intervals for Estimation
- Visualizing Confidence Intervals
- Using `scipy.stats` for statistical calculations

9. Time Series Analysis (if applicable)

9.1 Time Series Data Preparation

- Handling DateTime Index
- Resampling and Aggregation
- Dealing with Missing Data in Time Series

9.2 Visualization of Time Series Data

- Line Plots and Area Plots
- Rolling Statistics (Moving Averages)
- Decomposing Time Series: Trend, Seasonality, Residuals

9.3 Time Series Analysis Techniques

- Autocorrelation and Partial Autocorrelation (ACF, PACF)
- Forecasting with ARIMA, SARIMA, Holt-Winters Method

10. Model Selection and Validation

10.1 Cross-Validation Techniques

- Understanding Cross-Validation: K-Fold, Stratified K-Fold
- Implementing Cross-Validation with Scikit-learn
- Understanding Bias-Variance Tradeoff

10.2 Model Evaluation Metrics

- Evaluating Regression Models: RMSE, MAE, R-squared
- Evaluating Classification Models: Accuracy, Precision, Recall, F1-Score, ROC Curve, AUC

11. Advanced Data Visualization (Optional)

11.1 Interactive Visualizations

- Using Plotly for Interactive Charts
- Dash by Plotly for Web-based Dashboards
- Building Interactive Dashboards for EDA

11.2 Advanced Visualizations with Seaborn and Matplotlib

- Customizing Visualizations (Titles, Labels, Legends)
- Heatmaps, Cluster Maps, and Pair Grids
- Animated Plots

12. EDA Projects and Case Studies

12.1 Beginner Projects

- Visualize Distribution of a Numeric Dataset (e.g., Age, Income)
- Perform a Correlation Study between Variables in a Dataset
- Clean and Preprocess a Real-World Dataset

12.2 Intermediate Projects

- EDA for a Kaggle Dataset (e.g., Titanic, House Prices)
- Create a Dashboard to Visualize Key Metrics of a Business
- Build a Classification Model for Predicting Customer Churn after EDA

12.3 Advanced Projects

- Perform EDA on Time Series Data (e.g., Stock Prices, Sales Forecasting)

- Analyze Complex Data (e.g., Image Data, Text Data) with Preprocessing and Visualization
- Build a Recommendation System after performing EDA on User-Item Interaction Data